



ENGINEERING MATHEMATICS-II

MATLAB

Department of Science and Humanities

Finding Jacobian:

If $u = xyz$, $v = y^2$, $w = x + z$, then evaluate the Jacobian J.

```
>> syms x y z
```

```
>> J=jacobian([x*y*z, y^2, x + z], [x, y, z])
```

```
>> d = det(J)
```

Out Put: $J = \begin{bmatrix} yz & xz & xy \\ 0 & 2y & 0 \\ 1 & 0 & 1 \end{bmatrix}$

$$d = 2*y^2*z - 2*x*y^2$$

Finding Jacobian Continuation:



If $u = x^2 - 2y$, $v = x + y + z$, $w = x - 2y + 3z$, then evaluate the Jacobian J.

```
>> syms x y z
```

```
>> J=jacobian([x^2-2*y, x+y+z, x-2*y+3*z], [x, y, z])
```

```
>> d = det(J)
```

Out Put: $j = [2*x, -2, 0] [1, 1, 1] [1, -2, 3]$

$$d = 10*x + 4$$

Finding Jacobian Continuation:

If $u = x^2 - 2y^2, v = 2x^2 - y^2$, where $x = r\cos\theta, y = r\sin\theta$, then evaluate the Jacobian J.

```
>> syms x y r theta
```

```
>> x=r*cos(theta);
```

```
>> y=r*sin(theta);
```

```
>> J=jacobian([x^2-2*y^2, 2*x^2-y^2 ], [r, theta])
```

```
>> d = det(J)
```

```
>> simplify(d)
```

Finding Jacobian Continuation:

Out Put:

$$j = [2*r*\cos(\theta)^2 - 4*r*\sin(\theta)^2, -6*r^2*\cos(\theta)*\sin(\theta)]$$

$$[4*r*\cos(\theta)^2 - 2*r*\sin(\theta)^2, -6*r^2*\cos(\theta)*\sin(\theta)]$$

d =

$$12*r^3*\cos(\theta)*\sin(\theta)^3 + 12*r^3*\cos(\theta)^3*\sin(\theta)$$

ans =

$$6*r^3*\sin(2*\theta)$$

Finding Jacobian Continuation:

If $u = x^2 - y^2$, $v = 2xy$, where $x = r\cos\theta$, $y = r\sin\theta$, then evaluate the Jacobian J.

```
>>syms x y r theta
```

```
>>x=r*cos(theta);
```

```
>>y=r*sin(theta);
```

```
>>j=jacobian([x^2-y^2, 2*x*y, ], [r, theta])
```

```
>>d = det(j)
```

```
>>simplify(d)
```

Finding Jacobian Continuation:



Out Put:

j =

$[2*r*\cos(\theta)^2 - 2*r*\sin(\theta)^2, -4*r^2*\cos(\theta)*\sin(\theta)]$

$[4*r*\cos(\theta)*\sin(\theta), 2*r^2*\cos(\theta)^2 - 2*r^2*\sin(\theta)^2]$

d =

$4*r^3*\cos(\theta)^4 + 4*r^3*\sin(\theta)^4 + 8*r^3*\cos(\theta)^2*\sin(\theta)^2$

ans = $4*r^3$

Finding Jacobian Continuation:

If $x = u(1 - v)$, $y = uv$, prove that $JJ' = 1$

```
>> syms u v
```

```
>> J=jacobian([u*(1-v), u*v,], [u, v,])
```

```
>> inv(J)
```

```
>> J*inv(J)
```

```
>> det(J*inv(J))
```

Finding Jacobian Continuation:

Out Put:

$$J = \begin{bmatrix} 1 - v & -u \\ v & u \end{bmatrix}$$

$$\text{ans} = \begin{bmatrix} 1 & 1 \\ -v/u & -(v - 1)/u \end{bmatrix}$$

$$\text{ans} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{ans} = 1$$

Evaluation of Double Integral :

Evaluate $\int_0^1 \int_0^{1-x} \frac{1}{(\sqrt{x+y})(1+x+y)^2} dy dx$

```
>> fun = @(x,y) 1./ ( sqrt(x + y) .* (1 + x + y).^2 ); (Here, 'dot' is necessary)
```

```
>> ymax = @(x) 1 - x;
```

```
>> q = integral2(fun,0,1,0,ymax)
```

Out put:

```
q = 0.2854
```

Evaluation of Double Integral Continuation:

Evaluate $\int_{-\pi}^{2\pi} \int_0^{\pi} [y \sin(x) + x \cos(y)] dy dx$.

```
>>fun = @(x,y) y.*sin(x)+x.*cos(y);
```

```
>>q = integral2(fun,-pi,2*pi,0,pi);
```

Out put:

q =

-9.8696

Evaluation of Double Integral Continuation:

Evaluate $\int_0^5 \int_0^{x^2} x(x^2 + y^2) dy dx$.

```
>>fun = @(x,y) x.*(x.^2+y.^2);
```

```
>>ymax = @(x) x.^2;
```

```
>>q = integral2(fun,0,5,0,ymax)
```

Out put:

```
q = 1.8880e+04
```

Evaluation of Double Integral Continuation:



Evaluate $\int_0^1 \int_0^{\sqrt{x}} xy dy dx$

```
>>fun = @(x,y) x.* y  
>>ymax = @(x) sqrt(x);  
>>q = integral2(fun,0,1,0,ymax)
```

Out put:

```
q =  
    0.1667
```

Evaluation of Double Integral Continuation:



$$\text{Evaluate } \int_0^{\pi/2} \int_0^{1/\sin\theta+\cos\theta} \frac{r}{\sqrt{r\cos\theta+r\sin\theta}((1+r\cos\theta+r\sin\theta)^2)} dr d\theta$$

```
>> polarfun = @(r, theta)
```

```
r./(sqrt(r.*cos(theta)+r.*sin(theta)).*((1+r.*cos(theta)+r.*sin(theta)).^2))
```

```
>>rmax = @(theta) 1./(sin(theta) + cos(theta))
```

```
>>q = integral2(polarfun,0,pi/2,0,rmax)
```

Out put:

```
q = 0.2385
```

Evaluation of Triple Integral:



Evaluate $\int_0^3 \int_0^2 \int_0^1 (x + y + z) dz dx dy$

```
>> fun = @(x,y,z) x+y+z
```

```
>> q = integral3(fun,0,3,0,2,0,1)
```

Output: $q = 18$

Evaluation of Triple Integral Continuation:



Evaluate $\int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{-\sqrt{1-x^2-y^2}}^{\sqrt{1-x^2-y^2}} (x \cos(y) + x^2 \cos(z)) dz dy dx$

```
>>fun = @(x,y,z) x.*cos(y) + x.^2.*cos(z)
>>xmin = -1;
>>xmax = 1;
>>ymin = @(x)-sqrt(1 - x.^2);
>>ymax = @(x) sqrt(1 - x.^2);
>>zmin = @(x,y)-sqrt(1 - x.^2 - y.^2);
>>zmax = @(x,y) sqrt(1 - x.^2 - y.^2);
>>q = integral3(fun,xmin,xmax,ymin,ymax,zmin,zmax)
```

Out put:

q = 0.7796

Evaluation of Triple Integral Continuation:



Evaluate $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x + y + z) dx dy dz$

```
>>fun = @(x,y,z) x+y+z  
>>xmin= 0;  
>>xmax = @(z) z;  
>>ymin = @(x,z) x-z;  
>>ymax = @(x,z) x+z;  
>>zmin = -1;  
>.>zmax = 1;  
>>q = integral3(fun,zmin,zmax,xmin,xmax,ymin,ymax)
```

Output: q = -1.1102e-16

Evaluation of Triple Integral Continuation:

Evaluate $\int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z} dx dy dz$. Given, $\log 2 = 0.3010$

```
>>fun = @(x,y,z) exp(x+y+z)
>>xmin = 0;
>>xmax = 0.3010;
>>ymin = 0;
>>ymax = @(x) x;
>>zmin = 0;
>>zmax = @(x,y) x+log(y);
q = integral3(fun,xmin,xmax,ymin,ymax,zmin,zmax)
```

Output: q = -0.0533



THANK YOU
